Pension Run

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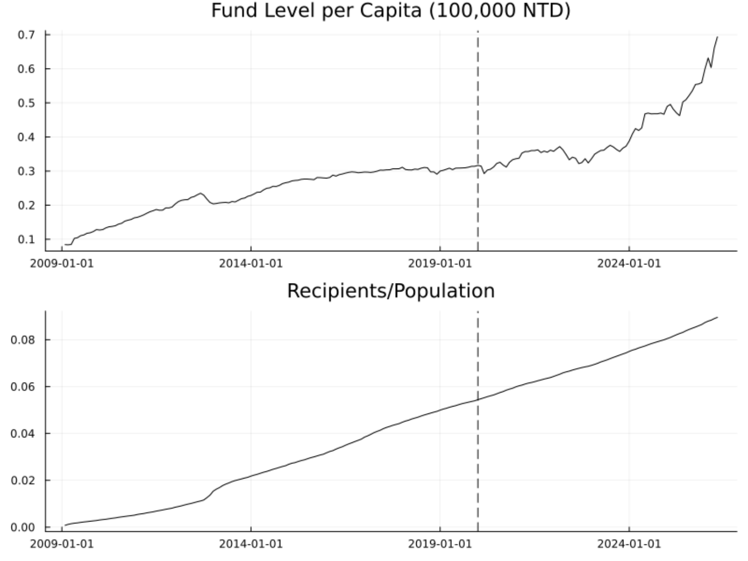
Motivation

- How does the potential reform of the pension system affect the retirement decisions?
  - Retire later to accumulate more wealth.
  - Retire earlier to withdraw more pension.
  - Reform Uncertainty.
  - Run on the pension system.
- Related literature:
  - **Pension reform**: Bi and Zubairy (2023); Daminato and Padula (2024); Delalibera, Ferreira, and Machado Parente (2026); French and Jones (2011); Kitao (2018).
  - **Bank run**: He and Xiong (2012); **?**.

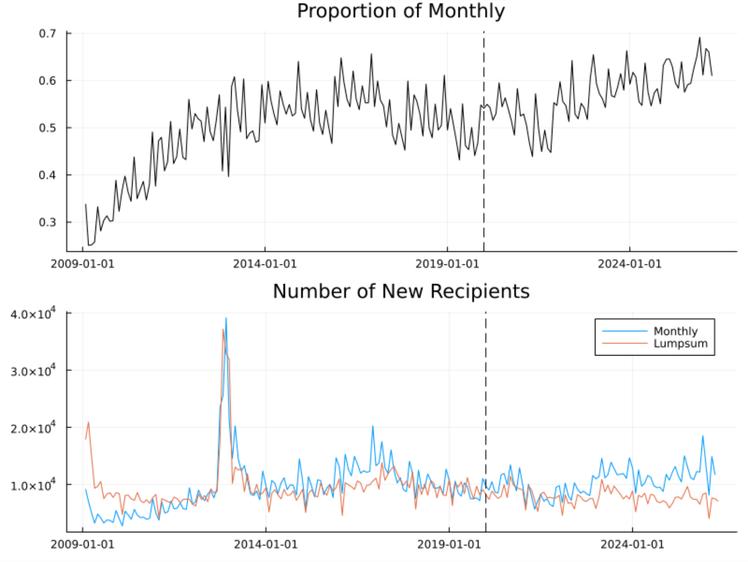
Background

- Since 2009, one with insurance records before 2009 can choose between lumpsum and monthly scheme.
- For a recipient with average monthly insured salary NTD $45,800$ with 30 years of working retired at standard retirement age and lives 12 years after retirement.
  - Lumpsum: $2,061,000$ NTD.
  - Monthly: $2,724,504$ NTD. (2% annual rate)
- Reform information shocks:
  - Reform commitee established in 2012.
  - Reform on Civil Servant Retirement Pension in 2017.

Data


Fund Level per Capita (100,000 NTD)
0.7
0.6
0.5
0.4
0.3
0.2
0.1
2009-01-01
2014-01-01
2019-01-01
2024-01-01
Recipients/Population
0.08
0.06
0.04
0.02
0.00
2009-01-01
2014-01-01
2019-01-01
2024-01-01

Data


Proportion of Monthly
0.7
0.6
0.5
0.4
0.3
2009-01-01
2014-01-01
2019-01-01
2024-01-01
Number of New Recipients
4.0×10^4
3.0×10^4
2.0×10^4
1.0×10^4
Monthly
Lumpsum
2009-01-01
2014-01-01
2019-01-01
2024-01-01

Model

Model Overview

- Continuous time and an infinite horizon.
- A continuum of workers chooses once, at retirement age T, between a lump-sum benefit ℓ and a monthly pension p.
- The pension fund earns a stochastic return and receives payroll tax revenue.
- Benefits stop when the fund balance reaches zero.
- Retirees' choices affect fund outflows and, consequently, the value of the monthly pension.

Population

Let $g(t,s)$ denote the population density at age s:

$$\frac{\partial g}{\partial t} = -\underbrace{\frac{\partial g}{\partial s}}_{\text{aging}} - \underbrace{m(s)g}_{\text{mortality}} + \underbrace{\int b(s)g(t,s)\,ds\,\delta_0}_{\text{birth}}.$$

Assume $b(s) = b$ and $m(s) = m\mathbf{1}\{s \geq T_m\}$. In a steady-age distribution, N_t grows at rate n and

$$\hat{g}(s) = \begin{cases} be^{-ns}, & s < T_m, \\ be^{-ns}e^{-m(s-T_m)}, & s \geq T_m, \end{cases}$$

where

$$1 = b\frac{1-e^{-nT_m}}{n} + be^{-nT_m}\frac{1}{n+m}.$$

Pension System

Let q_t be the share of new retirees choosing the monthly pension and Q_t the stock of monthly-benefit recipients. For $T \geq T_m$,

$$dQ_t = \Bigg[\underbrace{q_t \hat{g}(T) N_t}_{\text{new retirees}} - \underbrace{m Q_t}_{\text{mortality}}\Bigg]\, dt.$$

The pension fund balance evolves according to

$$dB_t = \Bigg[\underbrace{rB_t}_{\text{interest}} + \underbrace{\tau y N_t \int_{T_w}^{T} \hat{g}(s)\, ds}_{\text{pension tax}} - \underbrace{(1-q_t)\ell N_t \hat{g}(T)}_{\text{lump sum}} - \underbrace{pQ_t}_{\text{monthly}}\Bigg] dt$$
$$+ \sigma B_t\, dZ_t.$$

The system is bankrupt when B_t first reaches zero.

Detrended State Dynamics

Define fund balance and recipients per capita by $\hat{B}_t = B_t/N_t$ and $\hat{Q}_t = Q_t/N_t$. Then

$$d\hat{B}_t = \left[\underbrace{(r-n)\hat{B}_t}_{\text{net return}} + \underbrace{\tau y \int_{T_w}^{T} \hat{g}(s)\,ds}_{\text{pension tax}} - \underbrace{(1-q_t)\hat{g}(T)\ell}_{\text{lump sum}} - \underbrace{p\hat{Q}_t}_{\text{monthly}}\right] dt + \sigma \hat{B}_t\, dZ_t,$$

$$d\hat{Q}_t = \left[\underbrace{q_t \hat{g}(T)}_{\text{new retirees}} - \underbrace{(m+n)\hat{Q}_t}_{\text{mortality}}\right] dt.$$

- More lump-sum claims reduce the fund immediately.
- More monthly claims raise future liabilities through $\hat{Q}_t$.

Individual Pension Choice

A retiring individual chooses $d \in \{\ell, p\}$ with utility

$$u_{it}(d) = \begin{cases} \alpha + \ell + \beta\epsilon_i(\ell), & d = \ell, \\ V(\hat{B}_t, \hat{Q}_t) + \beta\epsilon_i(p), & d = p. \end{cases}$$

The value of monthly benefits is

$$V(\hat{B}, \hat{Q}) = \mathbb{E}\left[\int_0^{T_{\hat{B}}} e^{-(\rho+m)s} p \, ds \,\middle|\, \hat{B}, \hat{Q} \right].$$

With i.i.d. standard Gumbel ϵ_i, the monthly-benefit recipient share is

$$q(\hat{B}, \hat{Q}) = \frac{1}{1 + \exp\left(\frac{\alpha + \ell - V(\hat{B}, \hat{Q})}{\beta}\right)}.$$

Equilibrium and Main Mechanisms

In a Markov equilibrium:

1. State dynamics are consistent with retirees' choice q.
2. The value V is consistent with the induced bankruptcy time.
3. Individual choices aggregate to $q(\hat{B}, \hat{Q})$.

Two Externalities

1. Low $\hat{Q}$ $\longrightarrow$ far bankruptcy time $\longrightarrow$ monthly dominates $\longrightarrow$ add stress on the pension fund.
2. High $\hat{Q}$ $\longrightarrow$ near bankruptcy time $\longrightarrow$ lumpsum dominates $\longrightarrow$ accelerate bankruptcy.

Value Function and Numerical Solution

The monthly-pension value solves the nonlinear HJB equation

$$(\rho + m)V = p + \mathcal{A}V, \qquad V(0, \hat{Q}) = 0,$$

where

$$\mathcal{A} = \mu_B \frac{\partial}{\partial \hat{B}} + \mu_Q \frac{\partial}{\partial \hat{Q}} + \frac{1}{2}\sigma^2 \hat{B}^2 \frac{\partial^2}{\partial \hat{B}^2}.$$

- Nonlinearity arises because μ_B and μ_Q depend on q, which depends on V.
- At a sufficiently large upper boundary, impose $\partial V / \partial \hat{B} \approx 0$.
- Solve by finite differences and successive iteration, equivalent to semi-implicit scheme with $dt = \infty$.

To Do

- Relax the mortality assumption.
- Belief process.
- Endogenous retirement.
- Regime switch in fund return.